



Cairo University - Faculty Of Engineering
Computer Engineering Department
CMP4007 Applied Data Science - Spring 2026



Project Proposal

Team 9

Eng. Eman Hosam

#	Name	ID	Section	Bench No.
1	Ahmed Hamdy	9220032	1	4
2	Ahmed Aladdin	9220065	1	5
3	Asmaa Mohamed	9220154	1	29
4	Shehab Khaled	9220393	1	38

Contents

1	Problem Definition & Business Context	3
1.1	Business Problem	3
1.2	Stakeholder	3
1.3	Problem Formulation	3
1.4	Justification for Classification	3
1.5	Business Impact	3
2	Datasets	3
2.1	Primary Source (Kaggle Dataset)	3
2.2	Secondary Source (Web Scraping)	3
2.3	Feature Documentation (Core Fields)	4
2.4	Engineered Features (Planned)	4
3	Data Acquisition & Integration	5
3.1	Data Acquisition	5
3.2	Data Integration Strategy	5
4	Target Variable	5
5	Data Validation Report	6
5.1	Dataset Overview	6
5.2	Completeness - Missing Values Analysis	6
5.3	Descriptive Statistics (Numeric Columns)	7
5.4	Consistency - Schema & Data Types	7
5.5	Uniqueness - Duplicate Detection	7
5.6	Accuracy - Range & Outlier Checks	7
5.7	Validity - Categorical Analysis	8
5.8	Distribution Checks	8
5.9	Data Cleaning Plan	8
5.10	Summary of Data Quality Issues	8

1 Problem Definition & Business Context

1.1 Business Problem

Used car dealerships and online marketplaces receive large volumes of vehicle listings and trade-ins. Manually assigning each vehicle into a pricing/value tier is time-consuming and inconsistent across staff and branches. This project aims to automate the classification of incoming used cars into value tiers using historical market data.

1.2 Stakeholder

- Primary: Used Car Dealerships (pricing and inventory teams)
- Secondary: Online Marketplaces (listing recommendations) and Insurance Underwriters (risk and valuation support)

1.3 Problem Formulation

This problem is formulated as a **classification task**, where the goal is to predict the **price/value tier** of a used car listing.

Tier Definition: Cars are grouped into **Budget**, **Mid-Range**, and **Luxury** tiers based on price.

1.4 Justification for Classification

- The target is a discrete set of tiers (Budget / Mid-Range / Luxury)
- Tier outputs support operational decisions (pricing, procurement, and marketing)
- Classification enables threshold-based policies and consistent segmentation

1.5 Business Impact

- Faster and more consistent inventory categorization
- Improved pricing strategy and margin optimization
- Better customer targeting (budget buyers vs. premium buyers)

2 Datasets

2.1 Primary Source (Kaggle Dataset)

Source: Kaggle — *Uncovering Factors That Affect Used Car Prices*

URL: <https://www.kaggle.com/datasets/thedevastator/uncovering-factors-that-affect-used-car-prices>

autos.csv

Main structured dataset (CSV). Each row represents a used car listing with attributes such as price, brand, model, registration year, mileage, and fuel type.

2.2 Secondary Source (Web Scraping)

Source: AutoScout24

URL: <https://www.autoscout24.com>

AutoScout24 Listings

Scraped used car listings to increase coverage and validate pricing patterns.

2.3 Feature Documentation (Core Fields)

The main features to be used in modelling (10+ fields):

#	Feature	Type	Description
1	seller	Categorical	seller type (business/private)
2	vehicleType	Categorical	body type
3	yearOfRegistration	Numeric	Year the car was registered
4	gearbox	Categorical	Type of gearbox (manual or automatic)
5	power	Numeric	Power of the car in Horse Power
6	brand	Categorical	Brand of the car
7	kilometers	Numeric	Kilometers the car has been driven
8	fuelType	Categorical	Type of fuel (e.g. diesel, petrol, etc.)

Note: After deriving the tier label from **price**, the raw **price** feature will not be used as an input feature to avoid target leakage

2.4 Engineered Features (Planned)

In addition to the core fields, we plan to engineer the following features:

- **vehicleAge** = $\text{current_year (2026)} - \text{yearOfRegistration}$
- **kmPerYear** = $\text{kilometers} / (\text{vehicleAge} + 1)$
- **isAutomatic** = $(\text{gearbox} == \text{"automatic"})$
- **is_vintage_classic** derived from vehicleAge & brand
- **power_to_tier_bin** derived from powerPS

3 Data Acquisition & Integration

3.1 Data Acquisition

- Kaggle dataset downloaded as CSV file(s) (e.g., `autos.csv`).
- AutoScout24 listings collected via web scraping using Python (e.g., `Scrapy` + `Scrapy-Splash`).

3.2 Data Integration Strategy

- **Vertical merge** the Kaggle dataset with scraped AutoScout24 listings by aligning both sources to a unified schema.
- Standardize fields (e.g., currency, mileage units, categorical labels) and resolve naming differences (e.g., `kilometers` vs `mileage`).
- Deduplicate near-identical listings using matching keys such as **brand**, **model**, **yearOfRegistration**, **kilometers**, and **power**.
- Merge strategy:
 - Union rows after schema alignment (preferred for coverage)
 - Optional record linkage to detect overlaps across sources
- Handle missing fields introduced by scraping or schema differences using imputation or "unknown" categories.

Challenges:

- Inconsistent naming and category sets across sources
- Translating the Kaggle dataset from German to English
- Currency and locale differences
- Missing or noisy scraped fields due to unstructured HTML
- Potential duplicates across sources

4 Target Variable

Target Definition

PRICE_TIER = value tier derived from listing price.

Tier Construction:

Tier	Label	Price Range	Business Meaning	Business Interpretation
0	Budget	$\leq \$5,000$	High-mileage or older cars	Entry-level pricing segment
1	Mid-Range	\$5,000 - \$15,000	Recent-model mid-size cars	Mainstream segment
2	Luxury	$> \$15,000$	Luxury, electric, or new cars	Premium segment

Class Distribution [62.90% Budget, 25.57% Mid-Range, 8.62% Luxury]

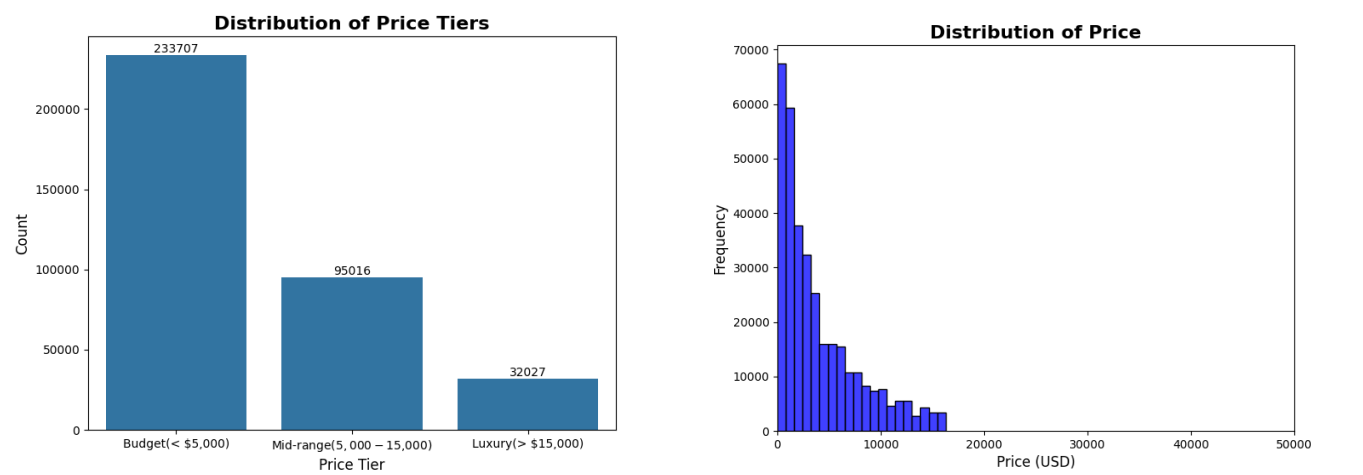


Figure 1: Left: Target tier counts. Right: Raw price histogram.

5 Data Validation Report

5.1 Dataset Overview

Metric	Value
Total rows (raw)	371528
Total columns	21
Numeric columns	8
Categorical columns	13
Total missing cells	194786
Full-row duplicates	0
Total missing cells	194786

5.2 Completeness - Missing Values Analysis

Column	Missing Count	Missing %	Status
vehicleType	37869	10.19%	WARN
fuelType	33386	8.99%	WARN
gearbox	20209	5.44%	OK
price_tier	10778	2.90%	OK

5.3 Descriptive Statistics (Numeric Columns)

Column	Count	Mean	Std	Min	25%	50%	75%	Max
yearOfRegistration	371528	2004.58	92.87	1000	1999	2003	2008	9999
powerPS	371528	115.55	192.14	0	70	105	150	20000
kilometer	371528	125618.69	40112.34	5000	125,000	150,000	150,000	150,000

5.4 Consistency - Schema & Data Types

Column	Detected Type	Expected Type	Match
dateCrawled	str	str	Yes
model	str	object	Yes
powerPS	int64	int64	Yes
kilometer	int64	int64	Yes
price	int64	int64	Yes
name	str	str	Yes
seller	str	str	Yes
offerType	str	str	Yes
yearOfRegistration	int64	int64	Yes
gearbox	str	str	Yes
price	int64	int64	Yes
vehicleType	str	str	Yes
abtest	str	str	Yes
monthOfRegistration	str	str	Yes
fuelType	str	str	Yes
brand	str	str	Yes
notRepairedDamage	str	str	Yes

All columns match expected schema with minor dtype differences are due to pandas representation.

5.5 Uniqueness - Duplicate Detection

- Full-row duplicates detected: **0 rows**
- Partial duplicates detected (removing dateCrawled column): **12575 rows**

5.6 Accuracy - Range & Outlier Checks

Column	Rule	Violation Percentage	Status
yearOfRegistration	[1900 - 2023]	182 (0.05%)	Pass
kilometer	[0 - 300,000]	0	Pass
powerPS	[40 - 5,000]	40907 (11.01%)	Fail
price	> 0	0	Pass

5.7 Validity - Categorical Analysis

Transmission Types:

- Automatic (73.81%), Manual (26.19%)

Fuel Types:

- Benzin (60.25%), Diesel (29.0%), lpg (1.45%), cng (0.15%), Electric (0.03%), Hybrid (0.07%)

5.8 Distribution Checks

Column	Mean	Median	Std	Skewness
registration year	2004.58	2003	92.87	72.13
price	14716	14861	6664	578.06
kilometers	125,618.69	150,000	40,112.34	-1.55
powerPS	115.55	105	192.14	58.2

Distributions fall within expected business ranges after excluding invalid values.

5.9 Data Cleaning Plan

- Remove invalid rows:
 - year outside [1900, 2023]
 - kilometers outside [0, 300000]
 - price is missing or price < 100
- Keep one record of the duplicated records
- Impute missing categorical values using mode value of the brand
- Standardize categorical labels across sources

5.10 Summary of Data Quality Issues

- Missing values in key variables (vehicleType, feulType, model, gearbox, price, notRepairedDamage).
- Class imbalance (ratio between Budget to Luxury ≈ 7.3)
- Potential outliers in **power**
- Data integration gaps (schema differences / scraping noise)